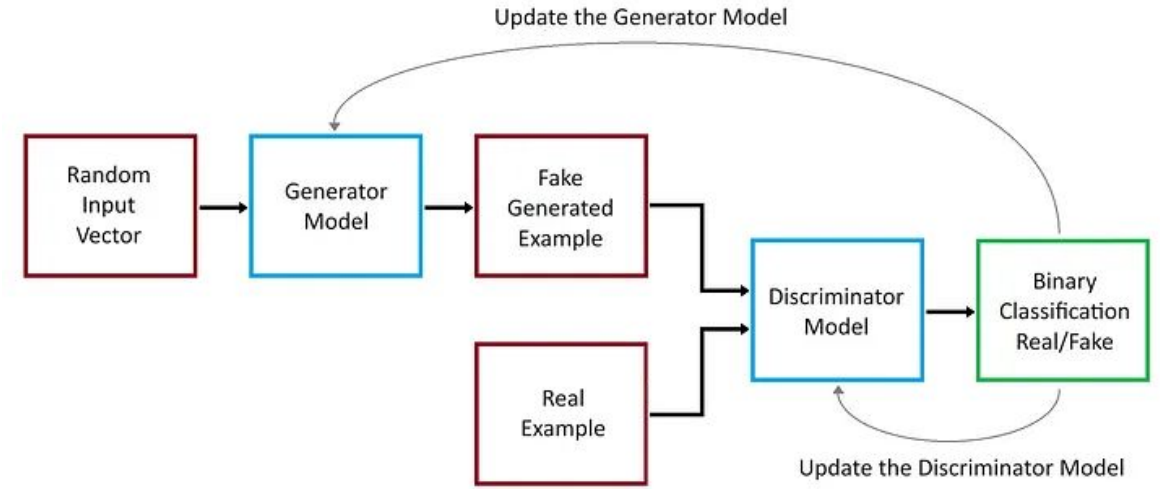
Week 6 Workshop

FIT5230 Malicious AI

Generative Adversarial Network

Generative Adversarial Network (GAN)

Image Source



Relationship between generator & discriminator **(competing in a game)**:

- the Generator aims to produce convincing fake data, and the Discriminator aims to tell real from fake.
- this adversarial process leads to the Generator creating increasingly better data over time.

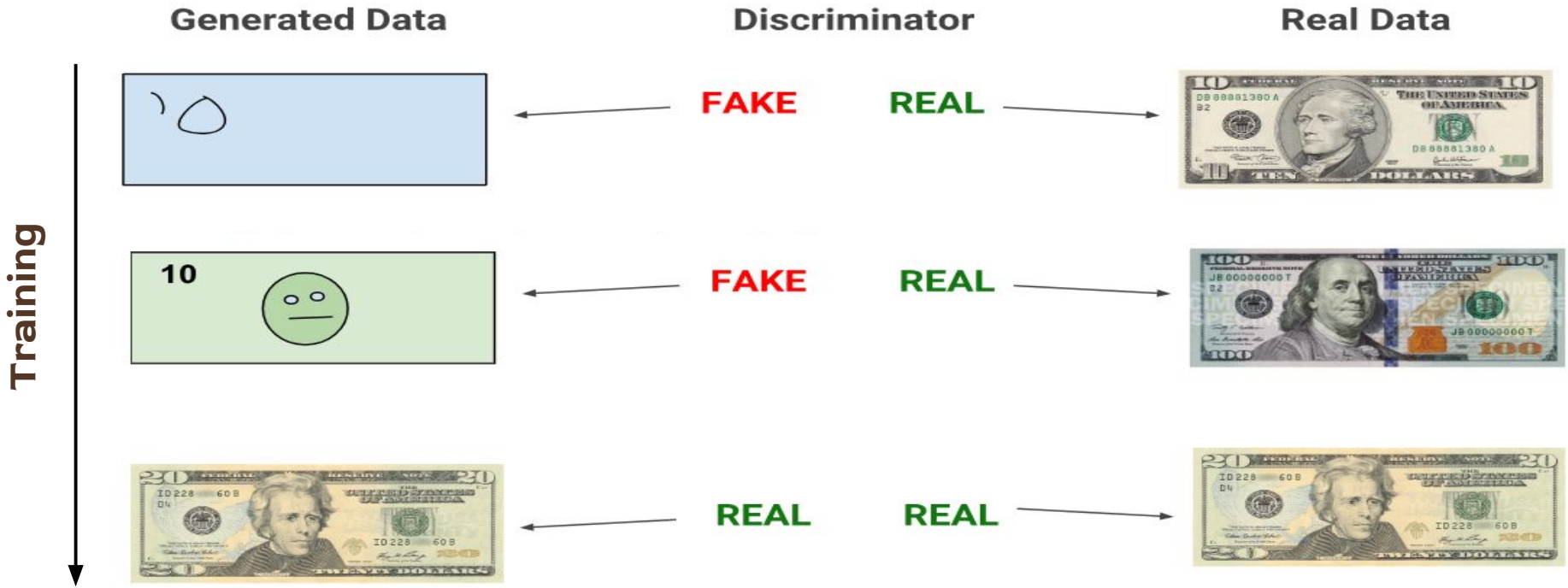


Image Source

Generator

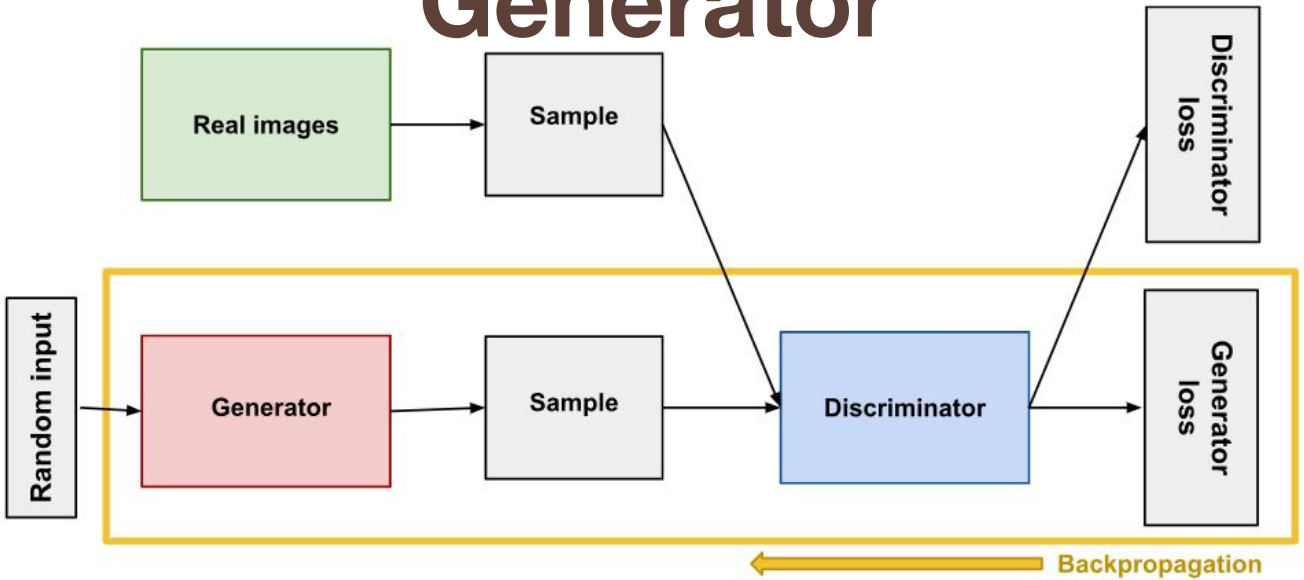
- Goal: **create fake data** by incorporating feedback from the discriminator and learns to make the discriminator classify its output as real.

- Input: The generator takes in a **random noise vector**, that serves as a seed or starting point for data generation → the generator starts without much knowledge about the true data distribution.

- Over time, as the GAN is trained, the generator learns to produce data that approximates the real data distribution.

Generator

Image Source



- Training process:
  - Sample random noise.
  - Produce generator output from sampled random noise.
  - Get discriminator "Real" or "Fake" classification for generator output.
  - Calculate loss from discriminator classification.
  - Backpropagate through both the discriminator and generator to obtain gradients.
  - Use gradients to change only the generator weights.

Generator

- The generator's goal is to produce samples that the discriminator incorrectly classifies as real.

- Its loss is typically represented as:

$$L_G = -\frac{1}{2}\mathbb{E}_{z\sim p_z(z)}[\log(D(G(z)))]$$

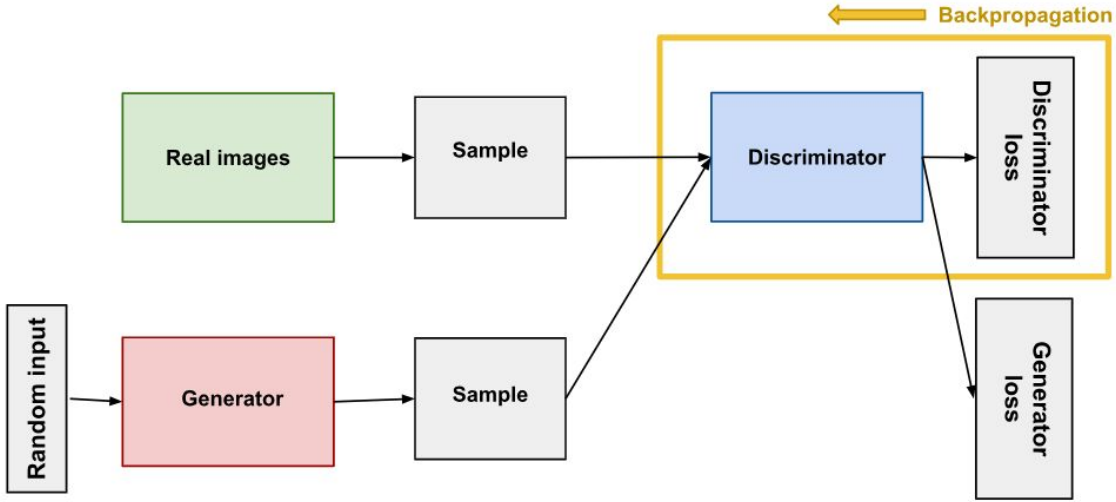
- The loss penalizes the generator when the discriminator correctly identifies its outputs as fake.

Discriminator

- Goal: **distinguish** real data from the data created by the generator.
- Training data:
  - **Real** data instances, such as real pictures of people → positive examples
  - **Fake** data instances created by the generator → negative examples
- **Input**:Receives data samples, which can be either real data or generated data.
- **Output**: Provides a scalar value between 0 (fake) and 1 (real), representing the probability that the input sample is real.

Discriminator

Image Source



- During discriminator **training**:
  - The discriminator classifies both real data and fake data from the generator.
  - The discriminator loss **penalizes** the discriminator for misclassifying a real instance as fake or a fake instance as real.
  - The discriminator **updates** its weights through **backpropagation** from the discriminator loss through the discriminator network.

Discriminator

- To correctly classify real samples as real and fake samples (produced by the generator) as fake.

$$L_D = -\frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}(x)}[\log(D(x))] - \frac{1}{2}\mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]$$

$$\mathbb{E}_{x\sim p_{\text{data}}(x)}[f(x)] \approx \frac{1}{N}\sum_{i=1}^{N} f(x_i)$$

(penalizes the discriminator for misclassifying real data)

$$\mathbb{E}_{z\sim p_z(z)}[f(z)] \approx \frac{1}{M}\sum_{i=1}^{M} f(z_i)$$

(penalizes the discriminator for misclassifying the fake data produced by the generator)

- *x_i* are samples from the real dataset
- *N* is the number of samples from the real dataset
- *z_i* are samples from the noise distribution
- *M* is the number of samples from the noise distribution.

GAN Loss

- The combined GAN Loss (minimax loss) is a combination of the discriminator and generator losses.

- It can be expressed as:

$$L_{\text{GAN}} = \min_G \max_D L_D + L_G$$

- The generator and the discriminator are in a two-player minimax game:

Discriminator	**Generator**
Ability to classify real/fake ↑	Ability to generate realistic data ↓
Ability to classify real/fake ↓	Ability to generate realistic data ↑

DCGAN @ICLR 2016

UNSUPERVISED REPRESENTATION LEARNING WITH DEEP CONVOLUTIONAL GENERATIVE ADVERSARIAL NETWORKS

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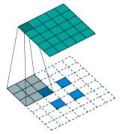
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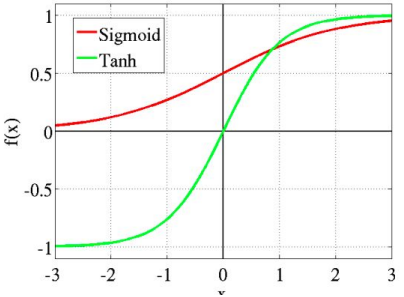
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ABSTRACT

In recent years, supervised learning with convolutional networks (CNNs) has seen huge adoption in computer vision applications. Comparatively, unsupervised learning with CNNs has received less attention. In this work we hope to help bridge the gap between the success of CNNs for supervised learning and unsupervised learning. We introduce a class of CNNs called deep convolutional generative adversarial networks (DCGANs), that have certain architectural constraints, and

DCGAN





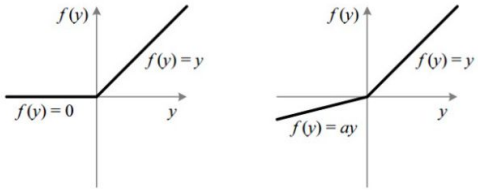
Generator (G)

- Input: Random noise vector (e.g., 100 values).
- Goal: Convert noise into a realistic image.
- Layers:
  - Uses transposed convolutions (also called deconvolutions) to upsample the noise into an image.
  - Applies ReLU activation in most layers.
  - Uses Tanh activation in the final layer to output pixel values between -1 and 1.
  - Includes batch normalization to stabilize training.

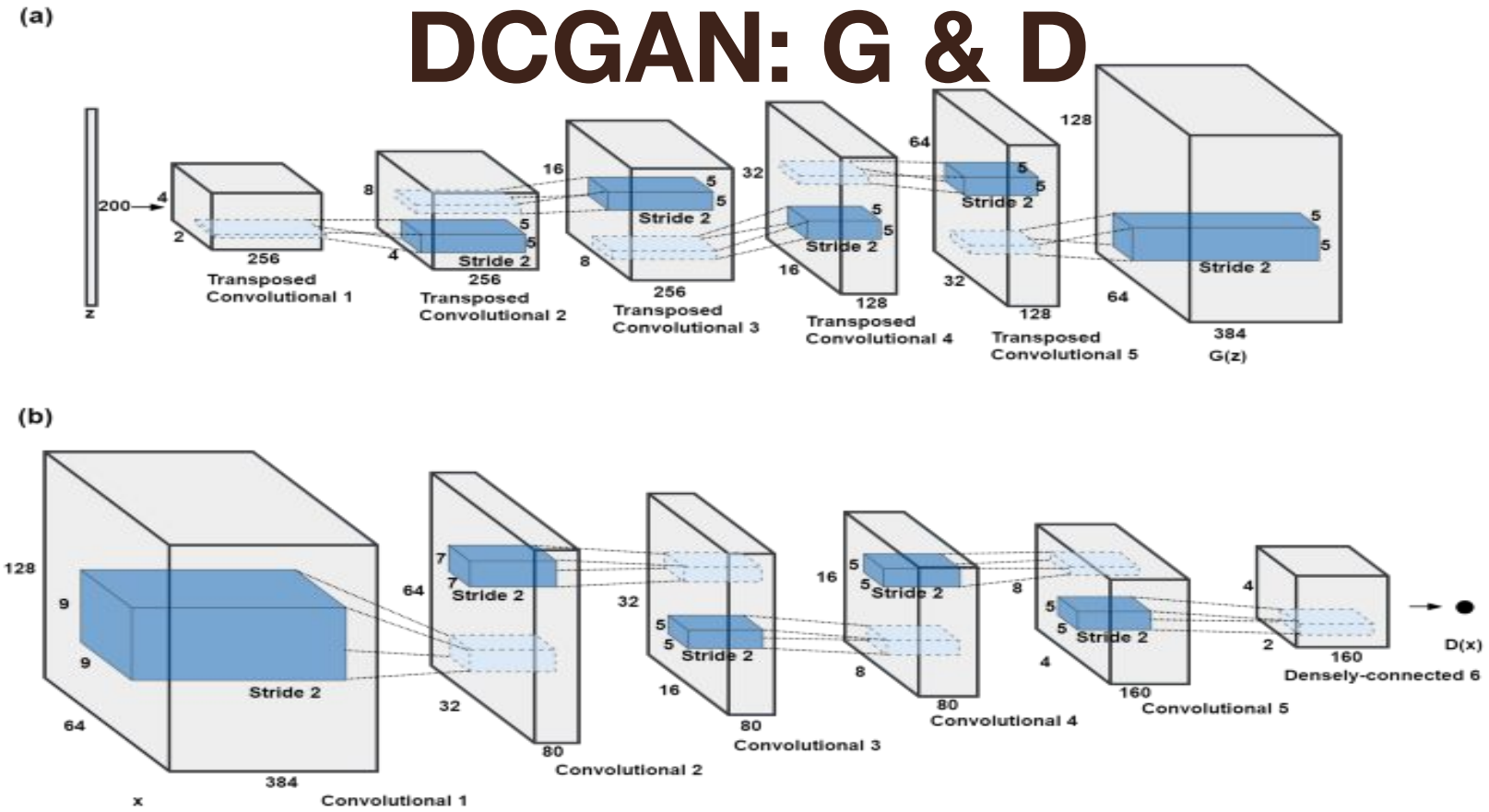
DCGAN

Discriminator (D)

- Input: An image (real or generated).
- Goal: Classify whether the image is real or fake.
- Layers:
  - Uses convolutions to downsample the image.
  - Applies LeakyReLU activation to allow small gradients for negative inputs.
  - Includes batch normalization for stability.
  - Outputs a single value between 0 and 1 (probability of being real).



DCGAN: G & D


(a)
200
z
4
2
256
Transposed Convolutional 1
8
4
Stride 2
256
Transposed Convolutional 2
16
8
5
Stride 2
256
Transposed Convolutional 3
32
16
Stride 2
128
Transposed Convolutional 4
64
32
Stride 2
128
Transposed Convolutional 5
128
64
Stride 2
384
G(z)
(b)
128
9
9
Stride 2
64
384
x
Convolutional 1
64
7
7
Stride 2
32
80
Convolutional 2
32
5
5
Stride 2
16
80
Convolutional 3
16
5
5
Stride 2
8
80
Convolutional 4
8
5
5
Stride 2
4
160
Convolutional 5
4
Stride 2
2
160
Densely-connected 6
D(x)

CycleGAN@ ICCV 2017

Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

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Abstract

Image-to-image translation is a class of vision and graphics problems where the goal is to learn the mapping between an input image and an output image using a training set of aligned image pairs. However, for many tasks, paired training data will not be available. We present an approach for learning to translate an image from a source domain X to a target domain Y in the absence of paired examples. Our goal is to learn a mapping $G : X \rightarrow Y$ such that the distribution of images from $G(X)$ is indistinguishable from the distribution Y using an adversarial loss. Because this mapping is highly under-constrained, we couple it with an inverse mapping $F : Y \rightarrow X$ and introduce a cycle consistency loss *to push $F(G(X)) \approx X$ (and vice versa). Qualitative results are presented on several tasks where paired training data does not exist, including collection style transfer, object transfiguration, season transfer, photo enhancement, etc. Quantitative comparisons against several prior methods demonstrate the superiority of our approach.*

cited > 28,000 times

CycleGAN

- Learns to translate between domains using **unpaired datasets**.
- CycleGAN consists of:
  - Two Generators:
    - G: X → Y (e.g., horse → zebra)
    - F: Y → X (e.g., zebra → horse)
  - Two Discriminators:
    - D_Y: distinguishes real vs fake images in domain Y
    - D_X: distinguishes real vs fake images in domain X

CycleGAN

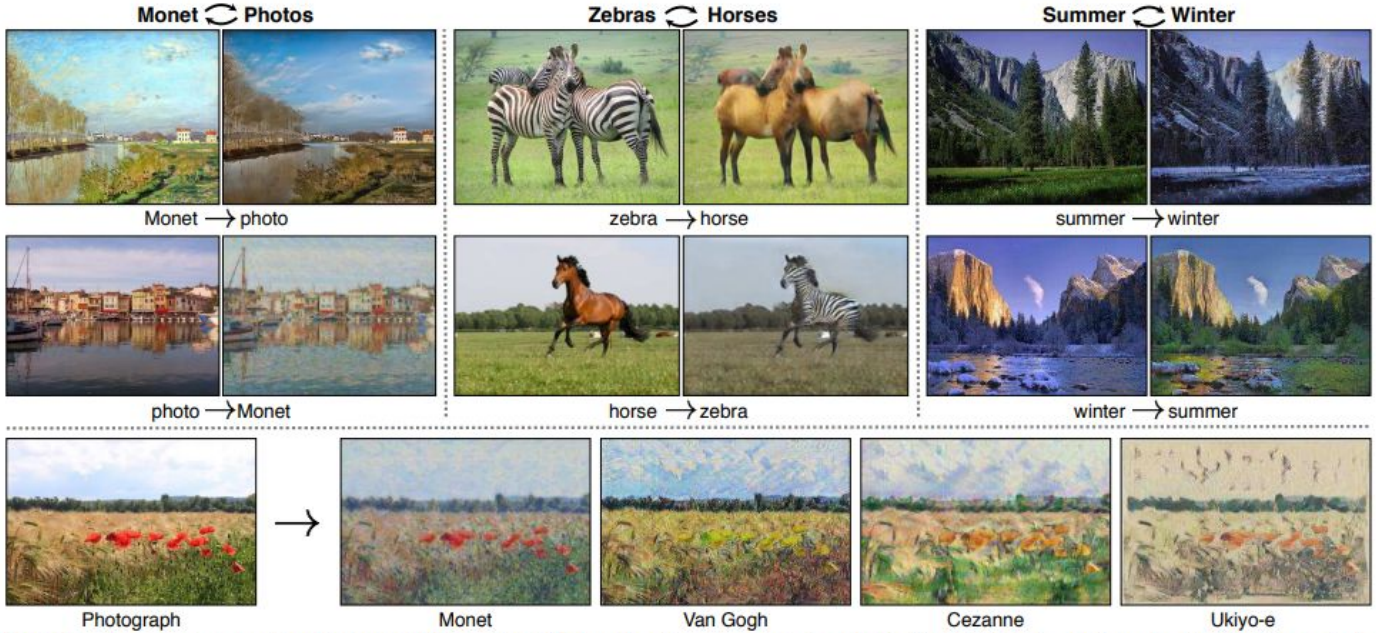
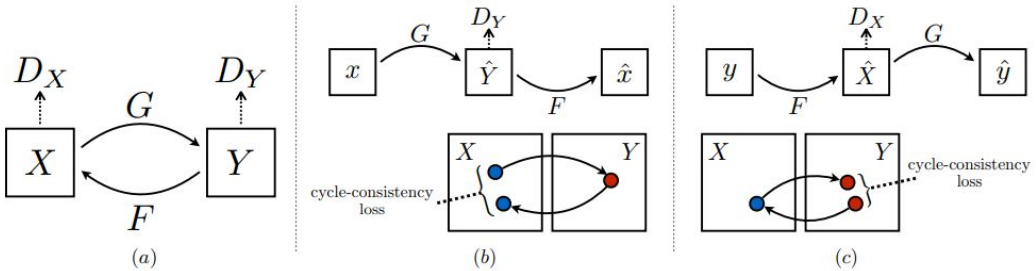


Figure 1: Given any two unordered image collections X and Y, our algorithm learns to automatically "translate" an image from one into the other and vice versa. Example application (*bottom*): using a collection of paintings of a famous artist, learn to render a user's photograph into their style.

CycleGAN

- Cycle consistency loss:
  - If you translate an image from domain *X* to *Y* using *G*, and then back to *X* using *F*, you should get the original image: $F(G(x)) \approx x$
  - Similarly: $G(F(y)) \approx y$
- Prevents the generators from producing arbitrary outputs and enforces semantic preservation.



CycleGAN

- Adversarial Loss encourages generators to produce realistic images that fool the discriminators:

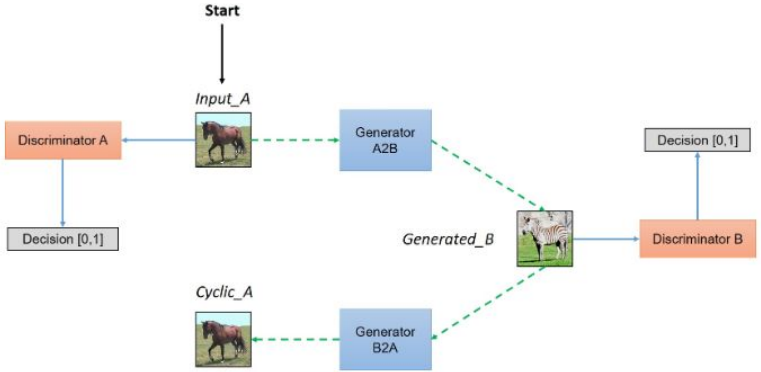
$$\begin{aligned}\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) =& \mathbb{E}_{y \sim p_{\text{data}}(y)}[\log D_Y(y)] \\ &+ \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log(1 - D_Y(G(x))],\end{aligned}$$

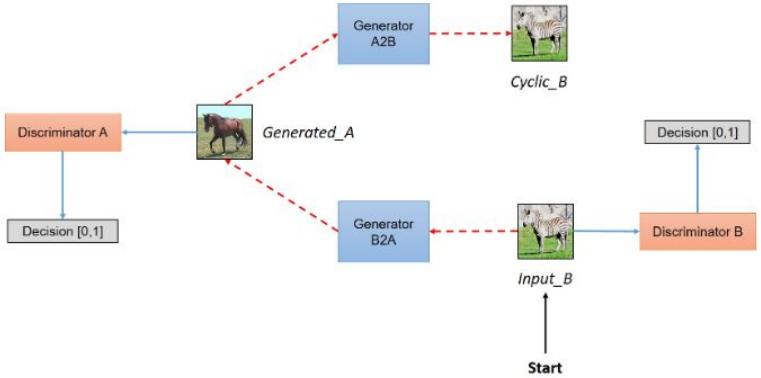
- Cycle Consistency Loss ensures that translating back and forth yields the original image:

$$\begin{aligned}\mathcal{L}_{\text{cyc}}(G, F) =& \mathbb{E}_{x \sim p_{\text{data}}(x)}[\|F(G(x)) - x\|_1] \\ &+ \mathbb{E}_{y \sim p_{\text{data}}(y)}[\|G(F(y)) - y\|_1].\end{aligned}$$

CycleGAN

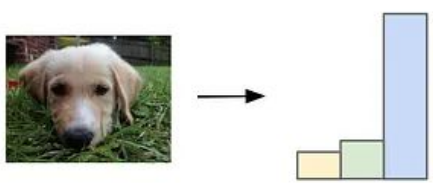
Network Architecture



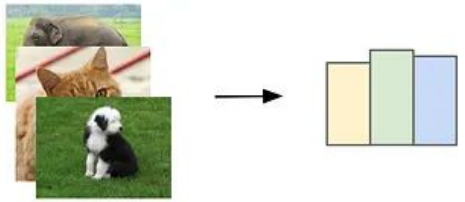


Evaluation Metrics for GANs

- **Inception Score (IS)**
  - Measures image quality and diversity.
    - Image Quality: A good image should be confidently classified into a specific class.
    - Image Diversity: The generated images should span across many different classes.
  - Uses a pretrained classifier (e.g., Inception v3).
  - High score = sharp, diverse images.



Ideal label distribution



Ideal marginal distribution

Image source

Inception Score (IS)*

- The Inception Score is defined as:

$$\exp(\mathbb{E}_{\boldsymbol{x}}\mathbf{KL}(p(y|\boldsymbol{x})||p(y)))$$

where:

- x be a generated image.
- p(y | x) be the conditional label distribution predicted by the Inception model.
- p(y) be the marginal distribution over all generated images.
- D_{KL} is the Kullback-Leibler divergence.
- E_x is the expectation over all generated images.

Image source

Inception Score (IS)*

- The Inception Score is defined as:

the probability of class labels given a generated image.

$$\exp(\mathbb{E}_{\boldsymbol{x}}\mathrm{KL}(p(y|\boldsymbol{x})||p(y)))$$

the average label distribution across all generated images

where:

- x be a generated image.
- p(y | x) be the conditional label distribution predicted by the Inception model.
- p(y) be the marginal distribution over all generated images.
- D_{KL} is the Kullback-Leibler divergence.
- E_x is the expectation over all generated images.

Image source

Evaluation Metrics for GANs

- **Fréchet Inception Distance (FID)**
  - Compares statistics (e.g., mean and covariance) of real vs generated images.
  - Lower FID = closer to real distribution.
  - More robust than IS for detecting mode collapse.

Fréchet Inception Distance (FID)*

- FID is computed as:

$$\|\boldsymbol{m} - \boldsymbol{m}_w\|_2^2 + \operatorname{Tr}\left(\boldsymbol{C} + \boldsymbol{C}_w - 2\left(\boldsymbol{C}\boldsymbol{C}_w\right)^{1/2}\right)$$

Where:

- $\|\boldsymbol{m} - \boldsymbol{m}_w\|_2^2$ measures the difference in means
- Tr is the trace of a matrix (sum of diagonal elements)
- $\left(\boldsymbol{C}\boldsymbol{C}_w\right)^{1/2}$ is the matrix square root of the product of covariances.

Evaluation Metrics for GANs

- **Perceptual Path Length (PPL)**
  - Measures smoothness of latent space interpolation
  - Reflects how well the generator maps small changes in latent vectors to perceptually smooth changes in the generated images.
  - Lower PPL = more consistent transitions

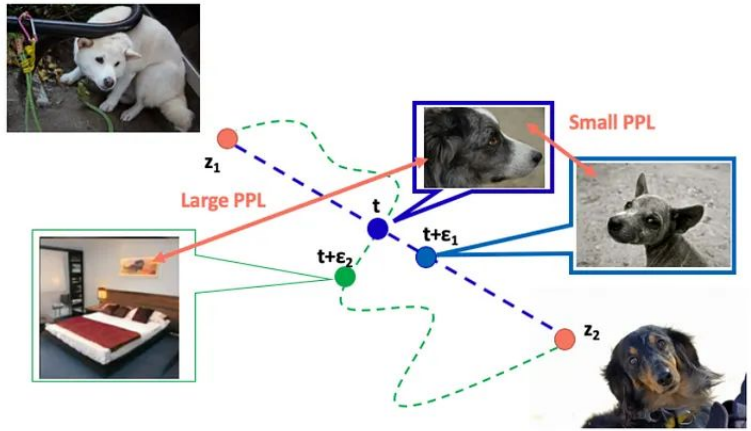


Image source

Perceptual Path Length (PPL)*

- PPL is computed as:
  - Sample two latent vectors z_1 and z_2
  - Interpolate between them using a small step ϵ: $z(t) = (1-t)z_1 + tz_2$
  - Generate images from interpolated latent vectors: $I(t) = G(z(t))$
  - Compute PPL:

$$\mathbb{E}_{z_1, z_2, t}\left[\frac{d(I(t), I(t+\epsilon))}{\epsilon^2}\right]$$

Where:

- d(x) is a perceptual distance (not pixel-wise)
- The expectation is taken over many random pairs and interpolation points

Evaluation Metrics for GANs

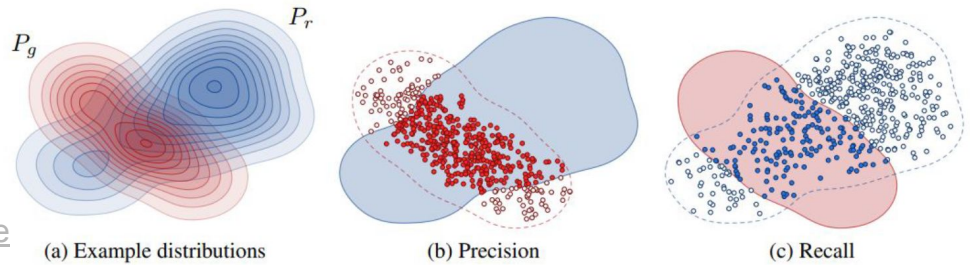
- **Precision** & **Recall**
  - Precision: How realistic are the generated samples?
  - Recall: How diverse are the generated samples?

- Embed images (real and generated) into a feature space using a pretrained network.
- Model the distributions of real and generated images in this space.
- For each generated image:
  - Check if it lies within the manifold of real images → contributes to precision.
- For each real image:
  - Check if it lies within the manifold of generated images → contributes to recall.

Precision & Recall

- Let:
  - G: set of generated samples
  - R: set of real samples
  - $\phi(x)$: feature embedding of image x
- Precision: Fraction of $\phi(g) \in G$ that are close to some $\phi(r) \in R$.
- Recall: Fraction of $\phi(r) \in R$ that are close to some $\phi(g) \in G$.
- “Closeness” is typically defined using a k-nearest neighbor or radius-based threshold in feature space.

Image source



Activity 1

What you need to do:

You are training a GAN to generate medical images (e.g., MRI brain scans) for data augmentation in a hospital setting. After several weeks of training, you obtain the following results from different evaluation metrics:

- Inception Score (IS): High
- Fréchet Inception Distance (FID): Moderate
- Perceptual Path Length (PPL): High
- Precision: High
- Recall: Low

1. What does each metric individually tell us about the GAN's performance in this scenario?
2. How would you interpret these results as a whole?
    a. Does the GAN generate convincing but limited (less diverse) images?
    b. Would these outputs be safe and reliable for use in medical augmentation?

Provide your answer here: https://shorturl.at/jG3lX